

HW3

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Overview of New Capabilities

The core of this extension is the integration of six pre-trained **Word2Vec models**, which provide the data for two new analysis views:

1. **Word2Vec Similarity**: A comparative table that shows words contextually similar to your search query across all six models.
2. **Vector Space (t-SNE)**: A set of six graphs that visually "map" the relationships between words, showing how closely related they are in the document set.

1. Word2Vec Similarity

- **Columns**: Each column represents one of the six Word2Vec models (e.g., "Skip-gram (Original)", "CBOW (Stemmed)").
- **Rows**: Each row is ranked from 1 to 10, with 1 being the most similar word.
- **Cell Content**: Each cell shows the similar word found by that model and a **Score**, which indicates the calculated degree of similarity (a higher score is better).
- **Stemmed Models**: The "Stemmed" columns offer unique insight. For a stemmed term like `treat`, they also show the original words found in the text (e.g., *treatment*, *treating*). This is excellent for understanding the root concepts related to your search.

2. Vector Space (t-SNE) View

The graphs are organized into three columns (**Original**, **No Stopwords**, **Stemmed**) and two rows (**Skip-gram**, **CBOW**). In each graph:

- **Large Red Point**: Represents your original search keyword.
- **Medium Blue Points**: The top 10 most similar words. You will see these points clustered near your red keyword.
- **Small Gray Points**: A random sample of other words from the documents. These provide a background context for the vector space.